

Introduction

- An attendance system that uses face recognition to attend students
- Mobile Application where teachers, students and admin and see and analyse the results
- Admins can manage classes and teachers

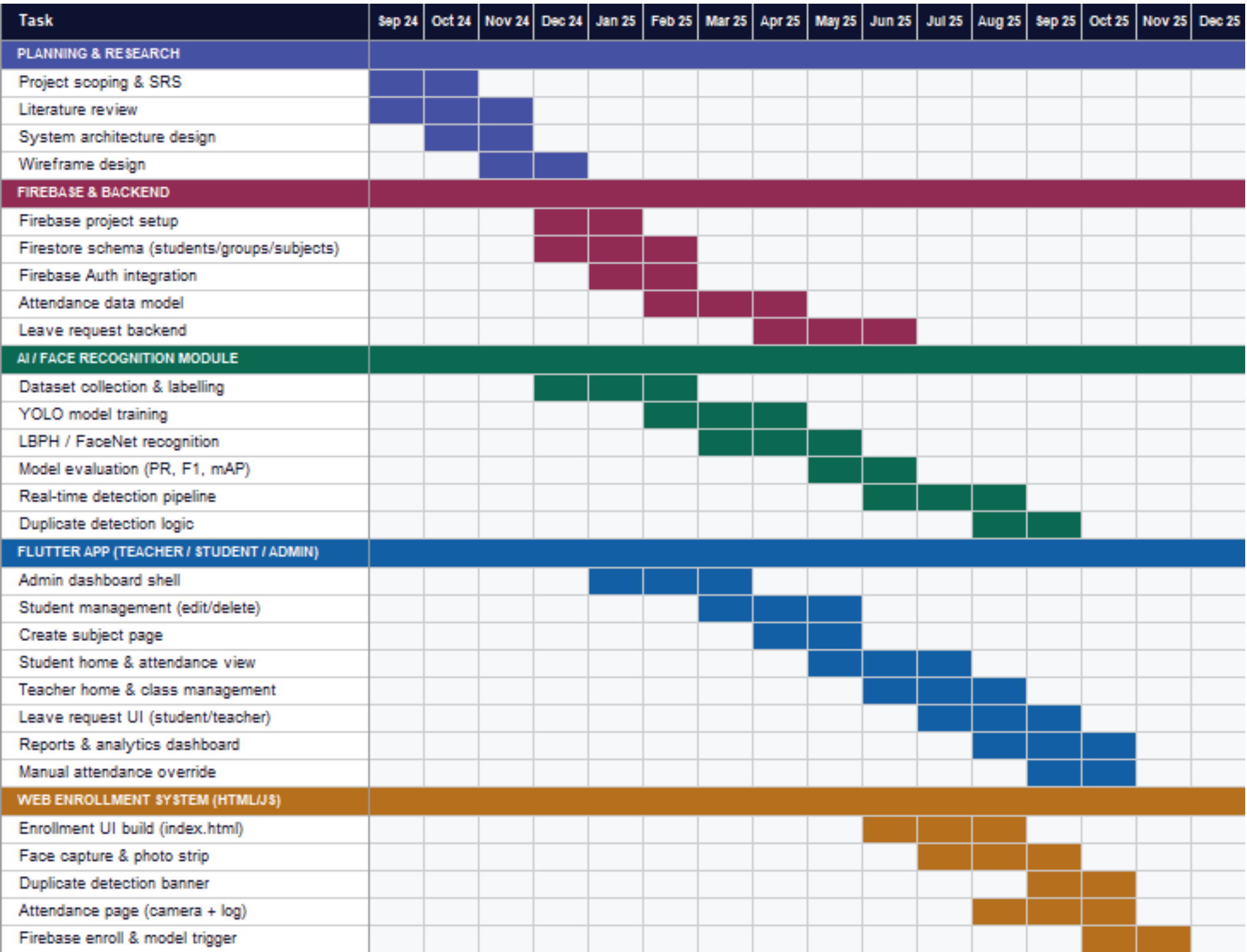
Academic Question

How can facial recognition technology be used to develop an efficient and automated attendance system?

Aims & Objectives

- To automate student attendance process using facial recognition
- To improve attendance accuracy and efficiency.
- To provide teachers with real-time insights and reports of students and their attendance
- To allow students and teachers to be notified about their classes.
- Provide a Secure cloud-based database integration to store the attendance.
- To avoid human errors and time consumptions during attendance.

Tracking Gantt Chart



Literature Review

- CNN generally offers superior accuracy and stability compared to LBPH (Budiman et al., 2023: p. 1, s. 1; Budiman et al., 2023: p. 6, s. 31). Thus YOLO is used for face detection which uses CNN.
- Nguyen et al. (2021: p. 534) employed Firebase for their attendance system implementation since it provides real-time database capabilities with automatic synchronization across clients.

Algorithm	Reported Accuracy	Reported Speed (FPS)	Source
Haar Cascade + LBPH	60% - 68%	10.8 - 16 FPS	(Winahyu et al., 2021: p. 1, s. 1; p. 4, s. 35)
CNN with face_recogniti on module	99.38%	Real-time	(Peddarapu et al., 2023)
YOLOv7	100%	5 FPS	(Lateef and Kamil, 2024: p. 43)

Model Evaluation

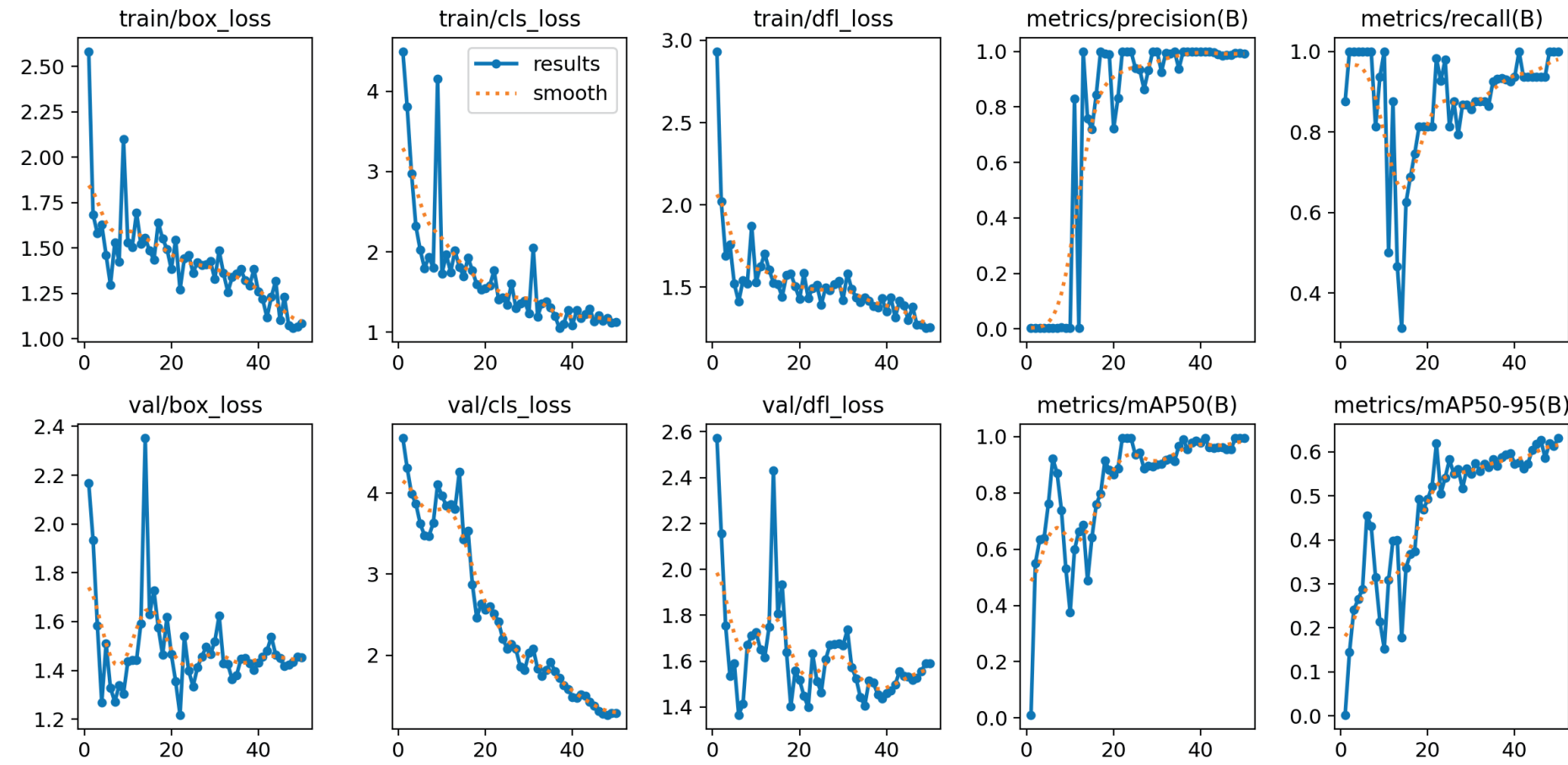
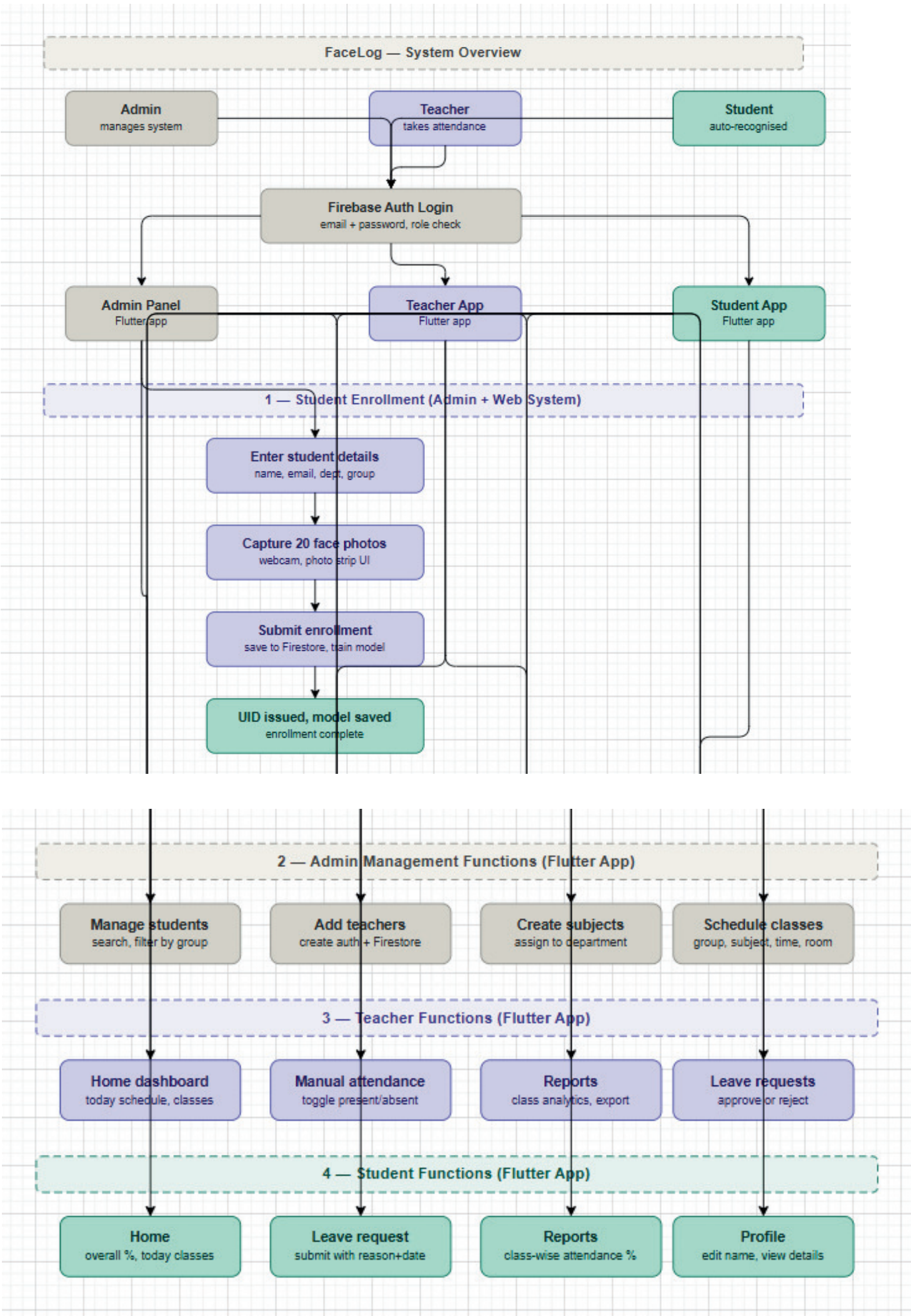


Fig: Model Evaluation of the face detection model

Tools Used



Product Process



Findings and Future Scope

- Offline mode: cache attendance locally and sync when reconnected
- Integration Into IOT: Use webcams for CCTV to automatically mark attendance.

Conclusion

- Automated attendance via facial recognition reduces manual effort
- Real-time Firestore sync keeps attendance data consistent across all apps
- The inclusion of features such as leave request management, analytics dashboards, and manual attendance override enhances the overall functionality and usability of the system for educational institutions.